WORKED SOLUTION

STEP III 1990, Question 10

Pure • Roots of equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q10

The key idea

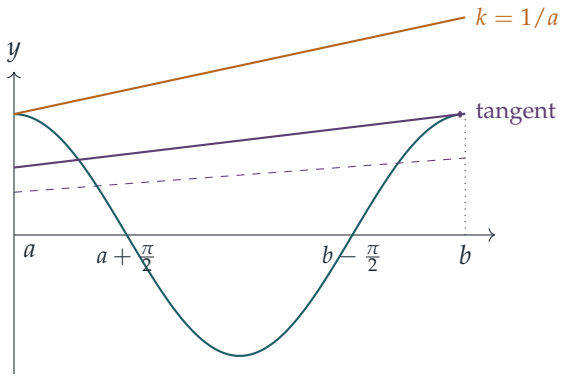
Over the given interval, $\sin x$ runs through one full period, starting and ending at a peak, while the line $y = kx$ is positive and climbs slowly. Roots can only occur where $\sin x > 0$, which happens on two separate stretches: one where $\sin x$ is falling (at most one crossing) and one where it is rising and concave (at most two crossings). The interesting borderline is when the line just touches the second arch, and writing the point of contact as a small distance δ to the left of the end of the interval turns the tangency condition into the given equation.

Solution

Throughout, write

$$a = (4n+1)\frac{\pi}{2}, \qquad b = (4n+5)\frac{\pi}{2} = a + 2\pi,$$

so the interval is $a < x < b$. Since $\sin a = \sin b = 1$, the graph of $\sin x$ starts at a peak, falls to 0 at $x = a + \frac{\pi}{2}$, is negative until $x = a + \frac{3\pi}{2} = b - \frac{\pi}{2}$, and then rises back to the peak at b.



The figure (drawn to scale for $n = 1$, with the origin far off to the left) shows $\sin x$ in teal, the line through $(a, 1)$ in amber, the tangent line in solid plum and a shallower line, which meets the curve three times, dashed.

The possible numbers of roots

Since $k > 0$ and $x > 0$, we have $kx > 0$, so every root lies where $\sin x > 0$: either in $I_1 = (a,\, a + \frac{\pi}{2})$ or in $I_2 = (b - \frac{\pi}{2},\, b)$.

On I_1. Here $\sin x$ is strictly decreasing and kx is strictly increasing, so $kx - \sin x$ is strictly increasing and there is at most one root. At the right-hand end $kx - \sin x > 0$; at $x = a$ it equals $ka - 1$. So there is exactly one root in I_1 if $ka < 1$, and none if $ka \geq 1$ (then $kx - \sin x > 0$ throughout).

On I_2. Let $f(x) = \sin x - kx$. Here $f''(x) = -\sin x < 0$, so f is concave and has at most two zeros on I_2. Also $f < 0$ at the left-hand end, and $f(b) = 1 - kb$.

- If $kb < 1$, then f goes from negative to positive, and a concave function can change sign only once on the way up, so there is exactly one root. The same holds when $kb = 1$: then $f(b) = 0$ and $f'(b) = \cos b - k = -k < 0$, so $f > 0$ just to the left of b, giving one root inside I_2 (the zero at b itself is excluded).
- If $kb > 1$, then $f < 0$ at both ends, so there are 0, 1 or 2 roots according as the maximum of f on I_2 is negative, zero or positive. The value 0 is the tangent case: the line touches the arch at some point $t \in I_2$, with slope k_t say. As k decreases from large values the line falls, so there are no roots for $k > k_t$, one for $k = k_t$ and two for $\frac{1}{b} < k < k_t$.

At the tangent point $k_t t = \sin t < 1$ and $t > a$, so $k_t < 1/a$; and $k_t > 1/b$ since the tangent case has $kb > 1$. Putting the two stretches together:

range of k	roots in I_1	roots in I_2	total
$k \leq 1/b$	1	1	2
$1/b < k < k_t$	1	2	3
$k = k_t$	1	1	2
$k_t < k < 1/a$	1	0	1
$k \geq 1/a$	0	0	0

So the equation may have 0, 1, 2 or 3 roots in the interval, as required.

Exactly one root

From the table, there is exactly one root precisely when $k_t < k < \dfrac{1}{a} = \dfrac{2}{(4n+1)\pi}$. It remains to describe k_t.

Ask yourself How do we turn "the line touches the curve" into equations?

The line $y = kx$ touches $y = \sin x$ at $x = t$ when the two graphs meet there *and* have

equal slopes:

$$\sin t = k_t t, \qquad \cos t = k_t.$$

The point t lies in I_2, so write $t = b - \delta$ with $0 < \delta < \frac{\pi}{2}$. Since $\sin b = 1$ and $\cos b = 0$,

$$\sin t = \sin(b-\delta) = \cos\delta, \qquad \cos t = \cos(b-\delta) = \sin\delta.$$

So the slope condition gives $k_t = \sin\delta$, and the meeting condition becomes

$$\cos\delta = \left((4n+5)\frac{\pi}{2} - \delta\right)\sin\delta.$$

Hence exactly one root occurs when

$$\sin\delta < k < \frac{2}{(4n+1)\pi},$$

with δ as described, which is what we had to show.

Approximations for large n

Divide the equation for δ by $\sin\delta$ (non-zero since $0 < \delta < \frac{\pi}{2}$):

$$\cot\delta = b - \delta, \qquad \text{that is} \qquad \tan\delta = \frac{1}{b-\delta}.$$

Since $\delta < \frac{\pi}{2}$, the right-hand side is less than $1/(b - \frac{\pi}{2})$, which is small when n is large. So δ is small, $\tan\delta \approx \delta$, and $b - \delta \approx b$, giving

$$\delta \approx \frac{1}{b} = \frac{2}{(4n+5)\pi}.$$

For the next approximation, write $\delta = \dfrac{1}{b} + \dfrac{c}{b^3}$ and keep terms up to b^{-3} on each side. On the left, $\tan\delta = \delta + \frac{1}{3}\delta^3 + \cdots = \dfrac{1}{b} + \dfrac{c + \frac{1}{3}}{b^3} + \cdots$. On the right,

$$\frac{1}{b-\delta} = \frac{1}{b}\left(1 - \frac{\delta}{b}\right)^{-1} = \frac{1}{b}\left(1 + \frac{\delta}{b} + \cdots\right) = \frac{1}{b} + \frac{1}{b^3} + \cdots,$$

using $\delta \approx 1/b$ in the correction term. Comparing the b^{-3} terms gives $c + \frac{1}{3} = 1$, so $c = \frac{2}{3}$ and

$$\delta \approx \frac{1}{b} + \frac{2}{3b^3} = \frac{2}{(4n+5)\pi} + \frac{16}{3(4n+5)^3\pi^3}.$$

(Equivalently, one step of the iteration $\delta \mapsto \arctan\dfrac{1}{b-\delta}$ starting from $\delta = 1/b$ gives $\arctan\dfrac{b}{b^2-1}$, which has the same expansion.) It follows that the lower end of the interval for k is $\sin\delta \approx \dfrac{1}{b} + \dfrac{1}{2b^3}$.

Checks

- For $n = 1$, solving $\cos\delta = (\frac{9\pi}{2} - \delta)\sin\delta$ numerically gives $\delta = 0.070973$, while $1/b = 0.070736$ and $1/b + 2/(3b^3) = 0.070971$: the improved approximation is about 150 times closer. For $n = 10$ the error falls to about 5×10^{-10}.
- Counting sign changes of $kx - \sin x$ on a fine grid for $n = 1, 2, 3, 5$ and a value of k from each row of the table gives $2, 3, 1, 0$ roots exactly as predicted, and random k between $\sin\delta$ and $2/((4n+1)\pi)$ always gives one root.

Common slips

- Forgetting the tangent case, or getting the order of $1/b$, k_t and $1/a$ wrong, which garbles the count.
- Using only $\tan\delta \approx \delta$ in the second approximation, which gives $\delta \approx b/(b^2 - 1) \approx 1/b + 1/b^3$. The $\frac{1}{3}\delta^3$ term in $\tan\delta$ is of the same order as the correction and must be kept.

Beyond STEP

Getting successively better approximations to the root of an equation with a small or large parameter is *asymptotic analysis*: you substitute a series in powers of $1/b$ and match coefficients, exactly as here. The same idea, applied to $\tan x = x$, describes the large roots of that equation, which turn up in heat-flow and vibration problems. For more on the analysis behind such expansions, see Bartle and Sherbert, *Introduction to Real Analysis*; Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

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